

SaaS Customer Churn & MRR Analysis

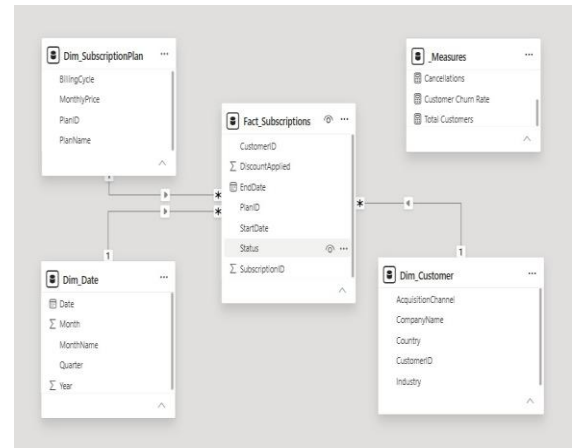
Power BI Data Engineering Project

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Executive Summary

End-to-end analytics pipeline tracking SaaS MRR and customer churn, featuring Python data generation, an optimised Star Schema, and advanced DAX time-intelligence logic.

Dashboard Overview & Model View



Data Architecture & Modelling

- Python Pipeline:** Simulated 1,000+ relational SaaS records using Pandas/Numpy to bypass flat CSV limitations.
- Power Query ETL:** Engineered a dynamic Dim_Date table via M-code and established clean data types.
- Star Schema:** Optimised relational model (1 Fact, 3 Dimension tables) for maximum VertiPaq performance.

Advanced DAX Logic

1. Baseline MRR (Dynamic Iterator)

Baseline MRR = SUMX(Fact_Subscriptions, RELATED(Dim_SubscriptionPlan[MonthlyPrice]) * (1 - Fact_Subscriptions[DiscountApplied]))

2. Active Customers (Time-Intelligence)

Active Customers = CALCULATE(DISTINCTCOUNT(Fact_Subscriptions[CustomerID]), REMOVEFILTERS(Dim_Date), Fact_Subscriptions[StartDate] <= MAX(Dim_Date[Date]), (Fact_Subscriptions[EndDate] > MAX(Dim_Date[Date]) || ISBLANK(Fact_Subscriptions[EndDate])))

3. Customer Churn Rate

Customer Churn Rate = DIVIDE([Cancellations], [Active Customers], 0)